MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

(E) TYPE 5AHP7A 1/

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Magnetic deflection, electrostatic focus, and aluminized screen

PIN CONNECTIONS AND DIMENSIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Ef	Ec1	Ec2	Eb2	Ehk	Rg1	Eb1	Alt
Unit:	V	Vdc	Vdc	Vdc	Vdc	Meg	Vdc	ft
Maximum:	6.9	0, -200	770	11,000	±180	(Note 1)	1,100	40,000
Minimum:	5.7	---	---	7,000	---	---	-500	---
TEST CONDITIONS:	6.3	Adjust	300	7,000	---	---	Focus	---

GENERAL:

Qualification - Required

1/ See note 5

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1002	Barometric pressure, reduced	-	Pressure - 140 ± 5 mmHg	---	---	---	---
1141	Pressure (implosion)	-		---	---	---	---
5111	Vibration	-		Width	---	2.0	mm
1331	Direct-interelectrode capacitance	-	k to all g1 to all g2 to all	Ck Cg1 Cg2	--- --- ---	8.0 10.0 10.0	pF pF pF
5201	Electrode current (grid No. 2)	-	Ec1 = 0	Ic2	-15	+15	μAdc
5216	Cathode illumination	-		---	---	---	---
(E) 1216	Base material insulating quality	-	Zone 5, minimum	---	---	---	---
	Quality conformance inspection, part 1						
5201	Voltage breakdown	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	-		---	---	---	---
5206	Gas ratio	4		Gr	---	0.25	---
5106	Bulb, screen, and faceplate quality	-		---	---	---	---
5231	Spot position (magnetic deflection)	-		---	---	7.94	mm
5236	Zero-bias anode current (magnetic deflection)	-		---	---	---	---
5241	Grid cutoff voltage	-		Ec1	-33	-77	Vdc
5251	Grid No. 1 leakage current	-		Ic1	---	3	μAdc
(E) 5246	Focusing electrode voltage	3	Ib2 = 100 μAdc	Eb1	0	250	Vdc
5223	Modulation	-	Ib2 = 100 μAdc, overscanned defocused pattern	ΔEc1	---	32	Vdc
	Quality conformance inspection, part 2						
1301	Heater current	-		If	540	660	mA
5101	Side terminal and base alinement	-	Pin position No. 5	---	---	---	---
5101	Neck and bulb alinement (magnetic types)	-		---	---	---	---
5101	Face tilt	-		---	---	---	---
5101	Neck straightness	-		---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued						
5216	Stray light emission (conventional types)	-	Eb2 = 11,000 Vdc; Ec2 = 770 Vdc; Eb1 = 0	---	---	---	---
5221	Screens	2		---	---	---	---
5226	Line width A (electrostatic deflection)	-	Ib2 = 100 μAdc	---	---	0.40	mm
5251	Heater-cathode leakage current	-		Ihk	---	10	μAdc
5251	Grid No. 2 leakage current	-		Ic2	---	5	μAdc
(E) 1111	Base pin solder depth	-		---	---	---	---
1101	Secureness of base, cap, or insert	-		---	---	---	---
1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	-	Group C; Eb2 = 11,000 Vdc; Ec2 = 770 Vdc; Eb1 at focus; t = 500 hours minimum, Ib2 = 2 μAdc	---	---	---	---
---	Life-test end points:						
5226	Line width A	-		Width	---	0.45	mm
5223	Modulation	-		ΔEc1	---	32	Vdc
5251	Heater-cathode leakage current	-		---	---	---	---
5251	Grid No. 1 leakage current	-		---	---	---	---
5251	Grid No. 2 leakage current	-		--	---	---	---
5216	Stray light emission (convention types)	-		---	---	---	--

NOTES:

1. When Ec2 is greater than 330 Vdc, Rg1 shall not exceed 0.5 Meg. when Ec2 is less than 330 Vdc Rg1 shall not exceed 1.5 Meg.
2. Anode voltage 7,000 Vdc, beam current 34 μAdc, limit is 430 cb minimum.
3. Using a 35 to 105 line raster with not less than a 1 4 inch (6.35 mm) spacing between adjacent lines, adjust for best focus at center of the tube.
4. This test to be performed at the conclusion of the holding period.
5. Tube types 5AHP4A, 5AHP7, and 5AHP19A have been deleted from this tube specification sheet.

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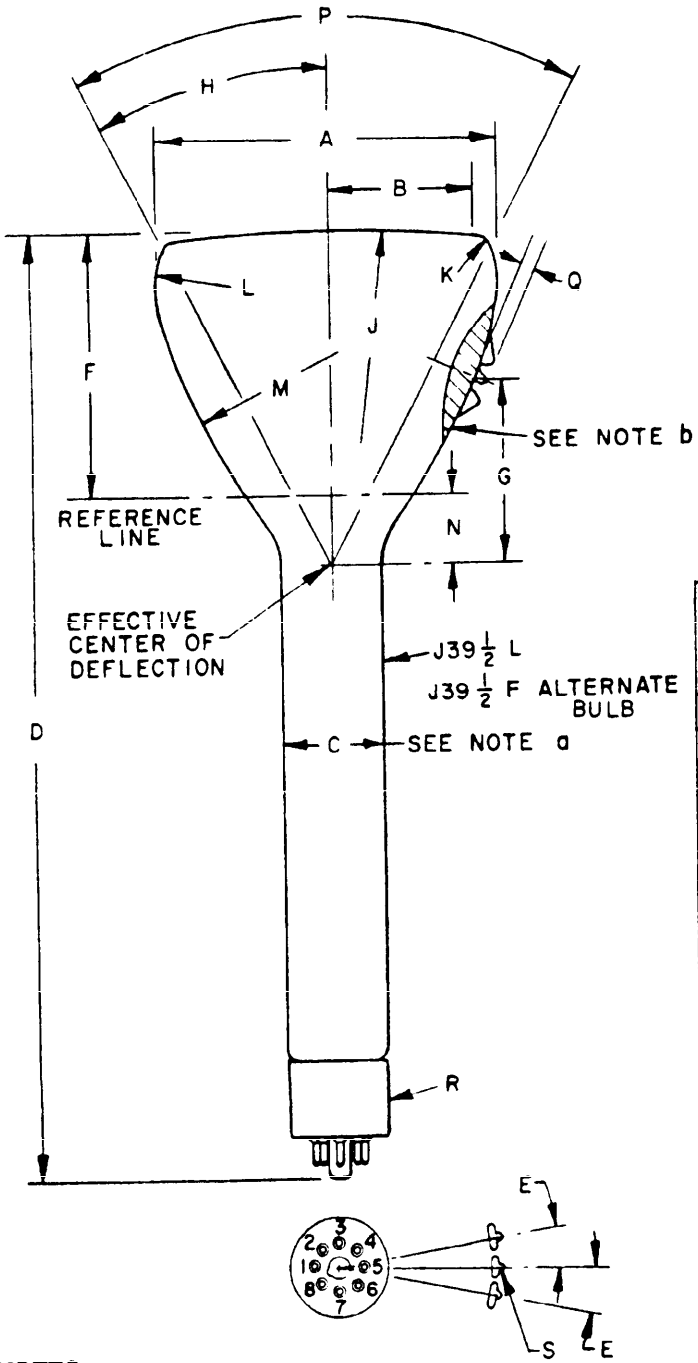
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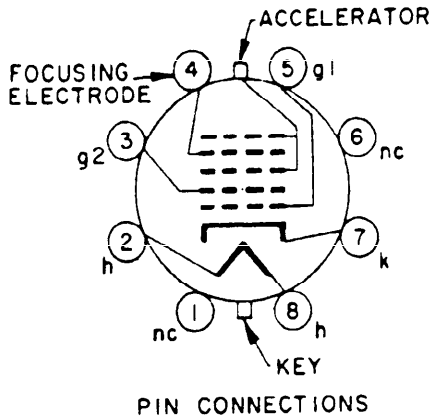
Preparing activity: Navy - EC

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PIN CONNECTIONS

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
R	Base: B8-11 or B8-65	
S	Bulb contact: J1-22	
Quality conformance inspection, part 2		
A	4.844(123.04)	5.031(127.79)
B	2.125 (53.98)R	
C	1.312 (33.32)	1.438 (36.53)
D	10.875(276.23)	11.375(288.93)
E		10°
F	3.500 (88.90)	3.750 (95.30)
G	1.188 (30.18)	1.563 (39.70)
Q		.188 (4.78)
Reference dimensions (see note c)		
H	26.50°	
J	24.000(609.60)R	
K	.125 (3.18)R	
L	1.125 (28.58)R	
M	10.000(254.00)R	
N	.950 (24.13)	
P	53°	

NOTES:
a. Dimension C applies to the entire length of the neck below the reference line. The reference line is determined by the point where a gage 1.430(36.24 mm) ±.003(.08 mm) inside diameter and 2 inches (50.80 mm) in length will stop against the bulb body.
b. Anti-Corona coating 1 min R concentric with contact. Do not handle tube by the part of the bulb having the anti-corona coating.
c. Reference dimensions are for information only and are not required for inspection purposes.

FIGURE 1. Outline drawing for electron tube type 5AHP7A.

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